EXP 10	Form Validation

Aim:

To validate the data entered in the Books and Book Issues forms using Deluge validation workflows.

Theory:

A validation workflow runs when the form is submitted. If a rule fails, alert shows a message and cancel submit stops the record from being saved.

- **Books rules:** Total Copies must be greater than zero, and Available Copies must be between 0 and Total Copies.
- **Issue rules:** The member must be Active, the book must have at least one available copy, a return date cannot be entered while issuing, and the due date cannot be before the issue date.

Procedure:

1. Go to Workflow > New Workflow, choose Books, record event Created or Edited, form event Validations on form submission, name it Validate Book Copies and add the script.
2. Create the workflow Validate Book Issue on Book Issues (Created, Validations on form submission) with the issue rules.
3. Test 1: add a book with Total Copies 0 and submit.
4. Test 2: issue a book to Rahul V (Inactive member).
5. Test 3: issue Astrophysics for People in a Hurry (1 copy, already issued to Sneha P) to Arun Kumar.

Program (Validate Book Copies - Books):

```
// Books - validation on form submission (Created or Edited)
if(input.Total_Copies == null || input.Total_Copies <= 0)
{
    alert "Total Copies must be greater than zero.";
    cancel submit;
}
if(input.Available_Copies != null && (input.Available_Copies < 0 ||
input.Available_Copies > input.Total_Copies))
{
    alert "Available Copies must be between 0 and Total Copies (" + input.Total_Copies
+ ").";
    cancel submit;
}
```

Program (Validate Book Issue - Book Issues, Created):

```
// Book Issues - validation when a book is issued (Created)
mem = Members[ID == input.Member];
if(mem.Member_Status != "Active")
{
    alert "Member " + mem.Full_Name + " is Inactive. Books can be issued only to active
members.";
    cancel submit;
}
```

```
bk = Books[ID == input.Book];
if(bk.Available_Copies == null || bk.Available_Copies <= 0)
{
    alert "No copies of '" + bk.Book_Title + "' are available. The book cannot be
issued.";
    cancel submit;
}
if(input.Return_Date != null)
{
    alert "Return Date cannot be entered while issuing a book.";
    cancel submit;
}
if(input.Due_Date != null && input.Due_Date < input.Issue_Date)
{
    alert "Due Date cannot be before the Issue Date.";
    cancel submit;
}
```

Output:

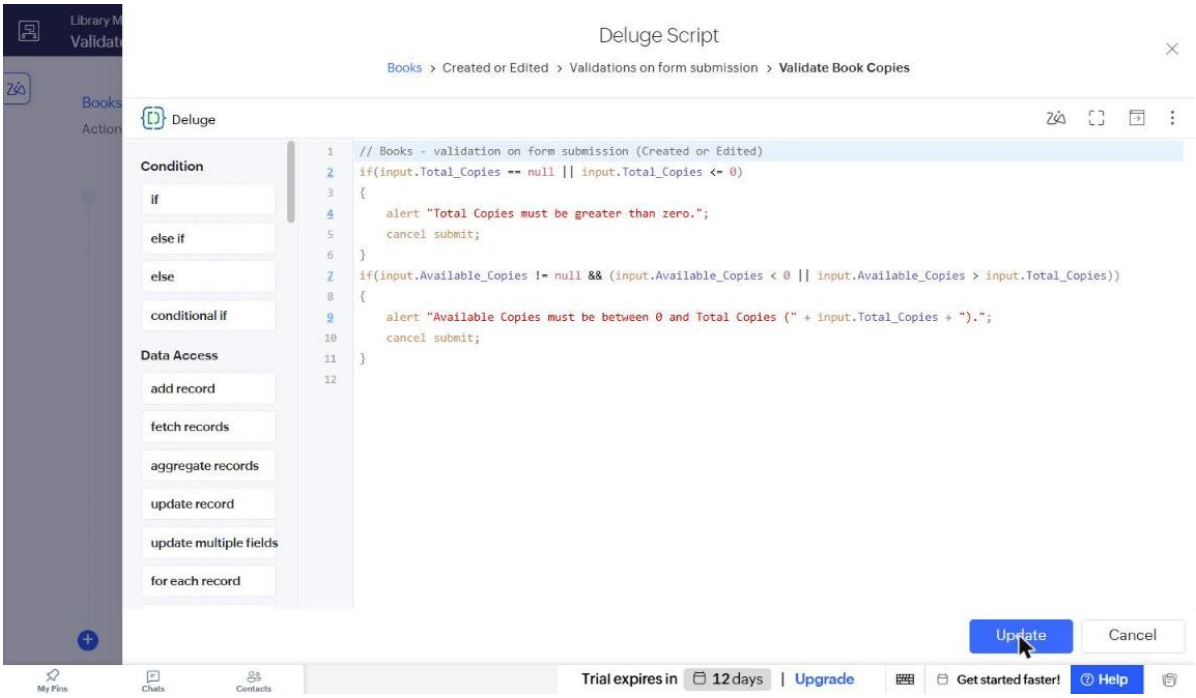


Fig. 10.1: Validate Book Copies script

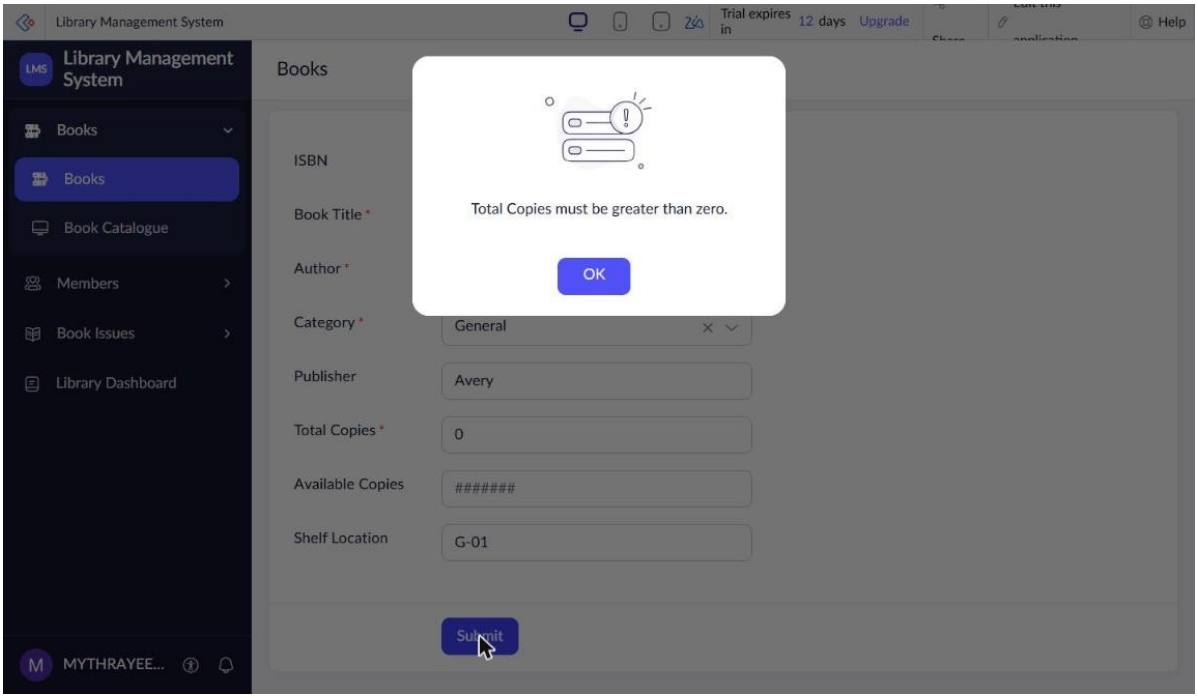


Fig. 10.2: Total Copies = 0 rejected

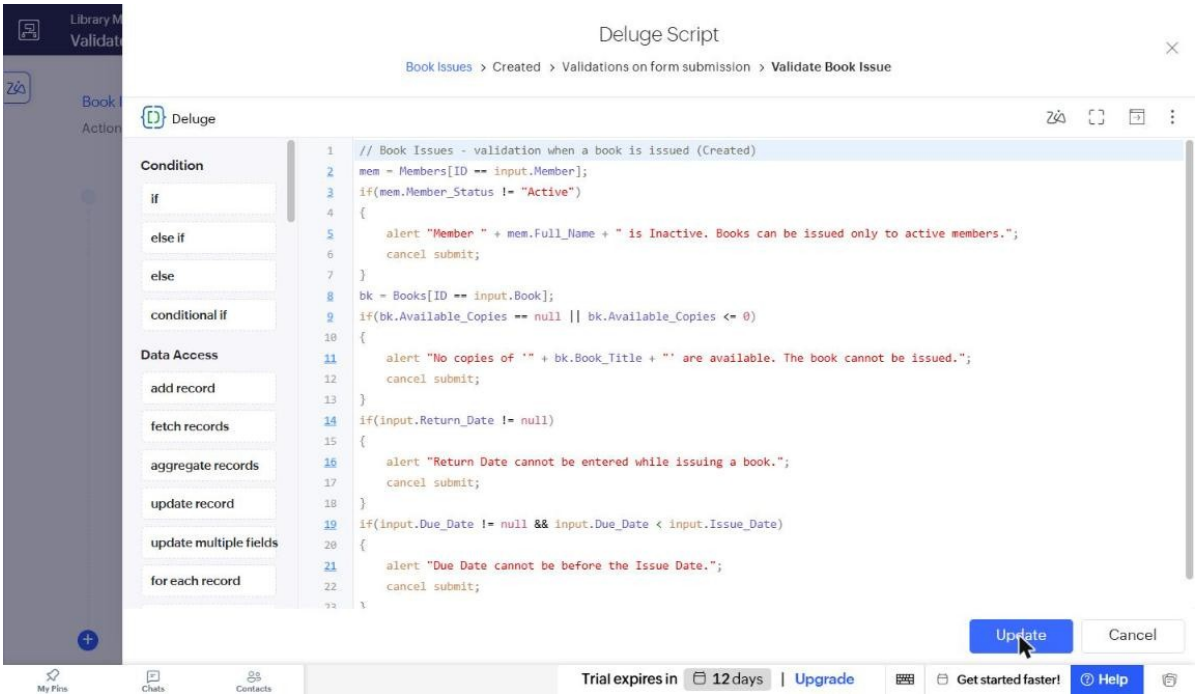


Fig. 10.3: Validate Book Issue script

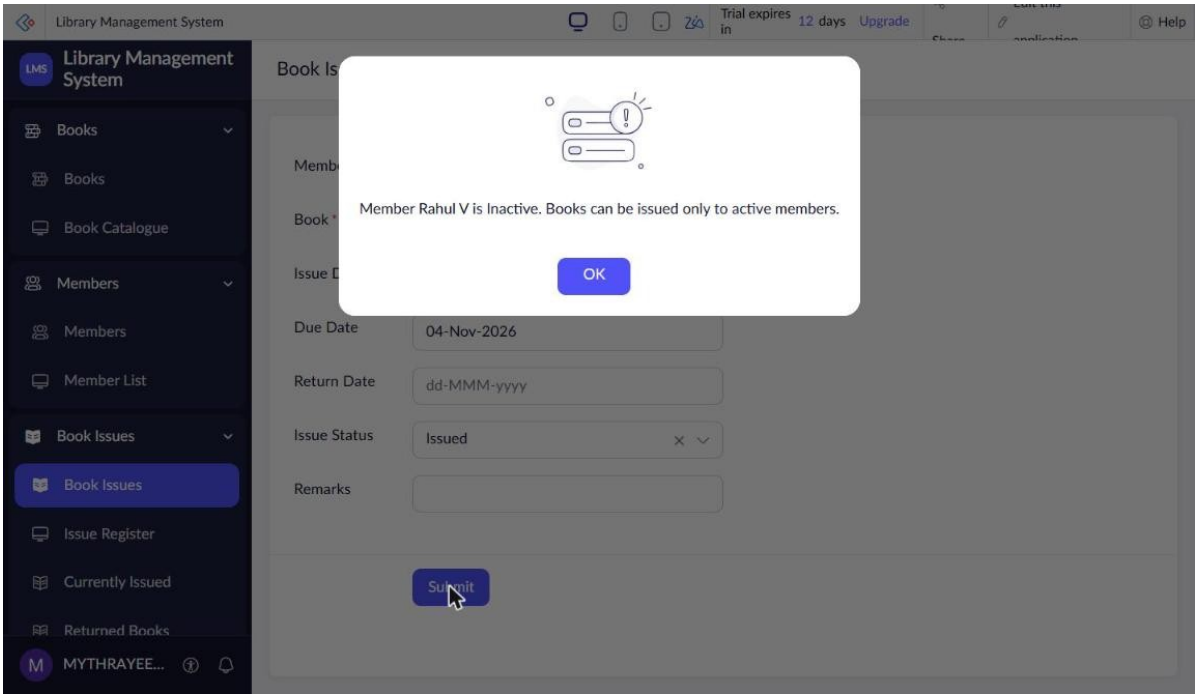


Fig. 10.4: Issue to inactive member blocked

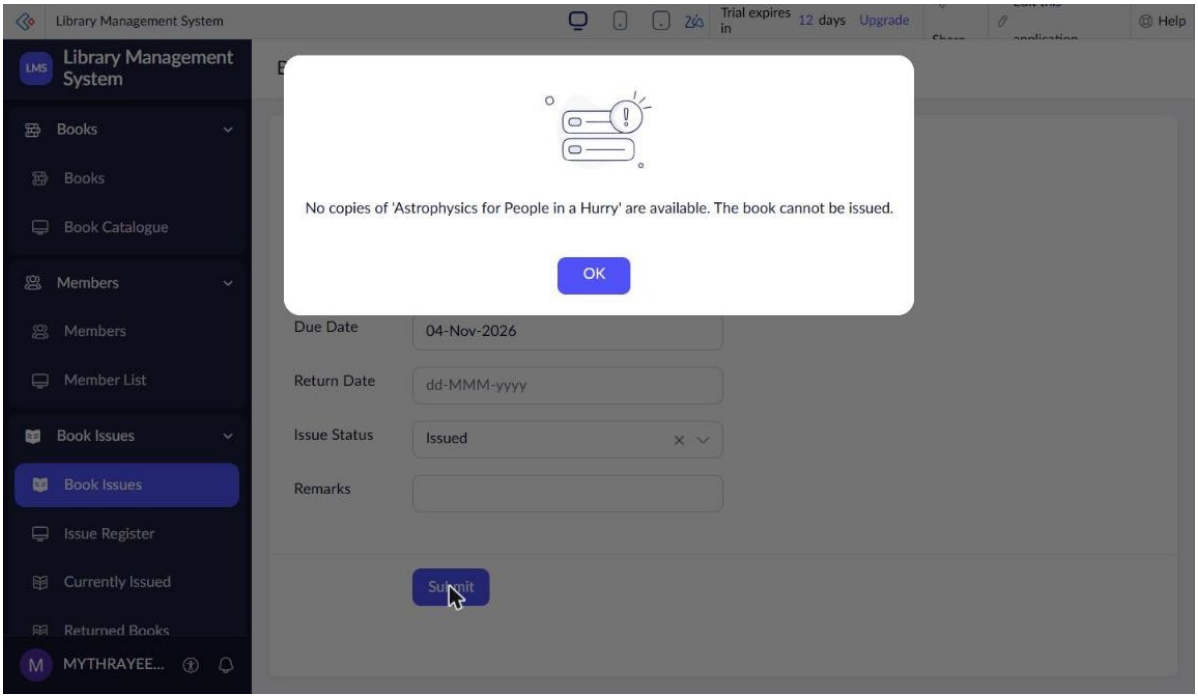


Fig. 10.5: Issue of a book with no available copies blocked

Observation:

Test	Input	Result
Zero copies	Atomic Habits, Total Copies 0	Blocked - "Total Copies must be greater than zero."
Inactive member	Rahul V - Introduction to Algorithms	Blocked - member is Inactive

No copies	Arun Kumar - Astrophysics (0 available)	Blocked - no copies available
Valid data	Atomic Habits, Total Copies 2	Saved, Available Copies set to 2

Result:

The validation workflows stopped invalid books and invalid issues and allowed valid records to be saved.